

OUTPUT-

1. Adam is working in an IT company. He has been given a task to reduce the load of a system by kitling some of the processes running in the LINUX operating system. Which commands will he use to complete the given task with the help of the following operation?
 - **Kill Processes by name**

```
root@LAPTOP-TUALCH80: ~  
ps -ef  
wsl: Failed to start the systemd user session for 'root'. See journalctl for more details.  
root@LAPTOP-TUALCH80:~#  
root@LAPTOP-TUALCH80:~# ps -ef  
UID          PID     PPID  C  STIME TTY          TIME CMD  
root           1         0  0  23:31 ?        00:00:01 /sbin/init  
root           2         1  0  23:31 ?        00:00:00 /init  
root           8         2  0  23:31 ?        00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate  
root        134         1  2  23:31 ?        00:00:00 /lib/systemd/systemd-journald  
root        181         1  4  23:32 ?        00:00:00 /lib/systemd/systemd-udev  
systemd+    208         1  1  23:32 ?        00:00:00 /lib/systemd/systemd-resolved  
systemd+    212         1  0  23:32 ?        00:00:00 /lib/systemd/systemd-timesyncd  
root        215         2  0  23:32 ?        00:00:00 /init  
root        216        215  0  23:32 ?        00:00:00 /init  
root        217        216  0  23:32 pts/0    00:00:00 -bash  
root        218         2  0  23:32 pts/1    00:00:00 /bin/login -f  
root        275         1  0  23:32 ?        00:00:00 /usr/sbin/cron -f -P  
message+   276         1  0  23:32 ?        00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only  
root        279         1  3  23:32 ?        00:00:00 /usr/bin/python3 /usr/bin/landscape-config --is-registered  
root        281         1  1  23:32 ?        00:00:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers  
syslog     282         1  0  23:32 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE  
root        283         1  1  23:32 ?        00:00:00 /usr/bin/snap wait system seed.loaded  
root        284         1  1  23:32 ?        00:00:00 /usr/lib/snapd/snapd  
root        285         1  0  23:32 ?        00:00:00 /lib/systemd/systemd-logind  
root        289         1  0  23:32 hvc0    00:00:00 /sbin/agetty -o -p -- \u --noclear --keep-baud console 115200,38400,9600 vt220  
root        296         1  0  23:32 tty1    00:00:00 /sbin/agetty -o -p -- \u --noclear tty1 linux  
root        323         1  1  23:32 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal  
root        324        218  0  23:32 pts/1    00:00:00 sh -c /usr/bin/env -i PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin run-parts --lsbysysinit /etc/  
root        325        324  0  23:32 pts/1    00:00:00 run-parts --lsbysysinit /etc/update-motd.d  
root        333        325  0  23:32 pts/1    00:00:00 /bin/sh /etc/update-motd.d/50-landscape-sysinfo  
root        344        333  0  23:32 pts/1    00:00:00 /bin/sh /etc/update-motd.d/50-landscape-sysinfo  
root        346        344  4  23:32 pts/1    00:00:00 /usr/bin/python3 /usr/bin/landscape-sysinfo  
root        408        181  0  23:32 ?        00:00:00 /lib/systemd/systemd-udev  
root        409        181  0  23:32 ?        00:00:00 /lib/systemd/systemd-udev  
root        442        217  0  23:32 pts/0    00:00:00 ps -ef  
root        443        284  0  23:32 ?        00:00:00 /usr/lib/snapd/snapd  
root@LAPTOP-TUALCH80:~# pkill color  
root@LAPTOP-TUALCH80:~# pkill colord  
root@LAPTOP-TUALCH80:~# pkill lxm1  
root@LAPTOP-TUALCH80:~# ps -ef|lxm1  
lxm1: command not found  
root@LAPTOP-TUALCH80:~# ps -ef
```

- **Kill a process based on the process name**

2. Write a program for process creation using C

- **Orphan Process**

```
root@LAPTOP-TUALCH80:~# nano orphan.c
root@LAPTOP-TUALCH80:~# gcc orphan.c -o orphan
./orphan
root@LAPTOP-TUALCH80:~# ./orphan
Parent Process exited
root@LAPTOP-TUALCH80:~# Child Process
PID = 1412
PPID = 270
```

- **Zombie Process**

```
nano zombie.c
root@LAPTOP-TUALCH80:~# gcc zombie.c -o zombie
zombie.c: In function 'main':
zombie.c:11:9: warning: implicit declaration of function 'system' [-Wimplicit-function-declaration]
   11 |         system("ps -l");
       |         ^~~~~~
root@LAPTOP-TUALCH80:~# ./zombie
Child Process exited
F S  UID    PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S   0      272     270  0  80   0 - 1530 do_wai pts/0    00:00:00 bash
0 S   0     1419     272  0  80   0 - 661 do_wai pts/0    00:00:00 zombie
1 Z   0     1420     1419  0  80   0 - 0 - pts/0    00:00:00 zombie <defunct>
0 S   0     1422     1419  0  80   0 - 723 do_wai pts/0    00:00:00 sh
4 R   0     1423     1422  0  80   0 - 1871 - pts/0    00:00:00 ps
```

3. Create the process using fork 0 system cgl' ‘

- **Child Process creation**
- **Parent Process creation**

```
root@LAPTOP-TUALCH80:~# nano fork.c
root@LAPTOP-TUALCH80:~# gcc fork.c -o fork
root@LAPTOP-TUALCH80:~# ./fork
Parent Process
PID = 1348
Child PID = 1349
Child Process
PID = 1349
PPID = 1348
root@LAPTOP-TUALCH80:~#
```

- **PPID and PID**

```
root@LAPTOP-TUALCH80:~# sleep 300 & [1] 2754
[1] 1443
[1]: command not found
root@LAPTOP-TUALCH80:~# ps -ef|grep sleep
root      1443      272  0 00:35 pts/0    00:00:00 sleep 300
root@LAPTOP-TUALCH80:~# kill 2754
-bash: kill: (2754) - No such process
root@LAPTOP-TUALCH80:~# killall sleep
[1]+  Terminated          sleep 300
root@LAPTOP-TUALCH80:~#
```